

Classification of Activities of Daily Living in Subjects with Parkinson's Disease using Artificial Neural Networks

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Abstract — Parkinson's disease (PD) is a progressive disease characterized by slow movements, tremors, and postural instability [1]. Inertial sensors provide valuable information to assess the quality of movement in patients with PD [2]. The present work proposes improving the follow-up and support systems issued by specialists in patients with PD by identifying and adequately classifying the activities of daily living (ADL) carried out by these patients by applying programmed artificial neuronal networks (ANN) in Python language. The proposed supervised learning method allowed the classification of 6 ADLs through 64 signals from inertial sensors that made up the database (*dataset*) to train a multilayer perceptron (MLP), achieving an accuracy of 99%. Also, by applying dimensionality reduction to *dataset*, it was possible to reach 80% of accuracy in the classification. These results can potentially help the specialist to reduce costs and achieve a more accurate evaluation, seeking to improve monitoring and quality in the rehabilitation of these patients.

Keywords — Parkinson's disease, Artificial Neuronal Networks, Inertial sensors.

I. INTRODUCTION

Parkinson's disease is a chronic and progressive process caused by neuronal degeneration in the substantia nigra and decreased dopamine values [3]. Treatment options have focused on replacing dopamine and providing symptomatic motor relief through drug therapy. However, this treatment becomes unsatisfactory after 4-6 years [4]. Due to the heterogeneity and complexity of PD symptoms, such clinical assessments can be challenging and time consuming. Ultimately, the only way to properly characterize a patient's motor status is to continuously evaluate their motor function over an extended period. Due to their small-size, light weight, and low-power, wearable motion sensors have already demonstrated their clinical relevance in healthcare and daily-life monitoring [5]. The evaluation of motor signs, such as tremors and secondary motor complications, such as fluctuations and dyskinesias, through inertial sensors, can provide reliable feedback on the effects of pharmacological, surgical, and/or rehabilitative therapeutic management [6]. Thus, the objective of this work is to classify, through the programming of ANN, the activities of daily living of patients with PD to help the specialists to a more accurate evaluation of motor functionality in treatment, seeking to improve monitoring and quality in the rehabilitation for these patients.

II. METHODOLOGY

Creation of the environment in Anaconda: As a first step, it was created a local environment dedicated to signal analysis in *Anaconda* that allows connection to the browser to create Jupyter notebooks for programming in Python language. For this, Anaconda Navigator (graphical user interface that allows to launch applications and manage *conda* packages, environments, and channels without using command line interface [7]) was downloaded. Then in the command console, the installation instructions were placed for the specialized libraries and frameworks to be used; such as *numpy*, *matplotlib*, *sci-kit-learn* and *Keras* that allowed the intelligent system to be implemented.

1) *Origin and Importing of data:* Once the environment was created, the database named 'Table2.mat' provided by the Luis Guillermo Ibarra Ibarra National Rehabilitation Institute (INR-LGII) was received. The database has 179,755 observations with 65 variables from signals in space-time and frequency from 4 inertial sensors such as average, minimum, and maximum values of accelerometers and triaxial gyroscopes located in the foot, thigh, pelvis, and wrist [8]. This database was renamed as 'dataset' where variables 1 to 64 are the inputs defined by the variable X, and vector 65 is the output Y; which has the types of ADLs carried out and labeled numerically from 1 to 6 (1=standing, 2=walking, 3=sitting, 4=climbing stairs, 5=lying down and 6=walking holding a tray in hand) by five subjects with Parkinson's disease with a mean age of 63.5 ±2.6 years and a mean score of 2 on the Hoehn-Yahr scale (patient with bilateral involvement without balance disturbance), which classifies to the patient according to the severity of the symptoms [6]. The three-dimensional components and the magnitudes of acceleration and angular velocity of the sensors were recorded at a sampling frequency of 250 Hz. The data from all five patients in the original database was preprocessed using a low-pass second-order Butterworth filter with a cutoff frequency of 5 Hz and smoothed using a moving average filter with a window size of 6.4 s. Motion data was previously used to train and test eight classification methods [9].

2) *Preprocessing of database:* Through programming in Python and specialized libraries, the following adjustment parameters were considered: First, normalization (Norm.) was carried out to a range between 0 and 1 for the input

values to increase the prediction performance of the algorithm used. On the other hand, having the outputs the output (types of activities) labeled numerically in decimal format from 1 to 6 can affect the learning performance of the network, so the classification was changed to One-hot encoding in binary numbers to avoid that one activity has more weight than another due to the number in its classification. Finally, supervised learning was performed by dividing the *dataset* 50%, using one half for the training stage and the other for testing.

3) *Neural network design*: As a first step, a first design called “Model A” was created in which the database was not randomized. In this model, three hidden layers were executed with the *ReLU* activation function (AF) were. The first layer had 32 neurons, the second 22, and in the last layer had 11 neurons. It was decided to use the *Adam* optimizer to improve the values of the parameters and to reduce the error made by the network. To improve learning, by reducing the probability of omitting information belonging to training and testing sectors, the *dataset* was later randomized. With this modification, “Model B” was obtained. Subsequently, the AF of the second layer was replaced in this model with *Sigmoid*, as same as an increase in the number of neurons was made, thus achieving “Model C”. By increasing the number of neurons, changing activation functions and number of layers, the efficiency of the models was increasing. Subsequently, following adjustments were applied to this third MLP model: a) the activation function *ReLU* was used in the input layer, b) in the following three hidden layers, the AF *Sigmoid* was declared, and c) a slight increase in the number of neurons in the hidden layers, except for the last one was made. This design was called “Model D,” and it obtained the best results in correctly recognizing ADLs. It was considered the best model due to the accuracy results to further tune the ANN and further improve its performance.

4) *Reduced database*: There is no method about the parameters to be measured, these are not routinely measured. So, to reduce cost (number of sensors), time (instrumentation time) and difficulty (problems of acquisition, processing and discomfort for users) it was decided to do a first test of reducing parameters through the use of Principal Component Analysis (PCA) and see its feasibility and continue with this work. Using 13 predictors previously obtained with PCA, which were considered as those variables that provide the most significant information in the database [10] it was decided to develop a new reduced database (*rDB*) where, instead of using 64 entries, only these 13 entries were used; with the purpose of creating a neural network model that develops a good learning without requiring much information. For this, from the original database, the variables (columns) 3, 6, 15, 21, 24, 31, 39, 40, 41, 47, 55, 56, 63 were taken and assigned as the new entries for the *rDB*; respecting column

65 as the output. Subsequently, the same conditioning process described in 3) was applied. As with the original base, 4 different network models were registered in this work where activation functions and combinations of parameters were tested to obtain the best possible percentage of accuracy. Starting with “Model A-13”, characterized by having only two hidden layers, not having randomized *rDB* and being normalized in the range of 0 to 1. This model was later decided to be modified its distribution % of the data, where 80% of the data was used for training and the other 20% for testing. With this modification “Model B-13” was reached; and to increase its accuracy percentage, we randomized the reduced database and added two hidden layers with *Sigmoid* AF and a smaller number of neurons in each layer, thus obtaining “Model C-13”. By changing the normalization range of this MLP to -1 to 1 we obtained the “Model D-13” which differs by having the Hyperbolic Tangent AF (*Tanh*) in the second, third and fourth hidden layer.

5) *Cross-validation*: To verify that the models performed had a reliable accuracy percentage, the statistical Cross-Validation technique was applied to “Model D” and “Model D-13” which were the ones that had the highest accuracy. This technique is used to evaluate Machine Learning models by training subsets of available input data to evaluate them with a complementary subset of data [11]. For this technique it was decided to randomize the data and have 10 folds with 100 epochs as parameters.

III. RESULTS

A. For dataset

Table 1 shows the characteristics and adjustment parameters applied to the main four models for *dataset*, as well as the accuracy achieved and shown graphically in Fig 1.

TABLE 1
PARAMETERS FOR MODELS A-D

Model	Norm.	Random	# layers	A.F.	# Neurons	Opt.	Accuracy %
A	0 a 1	No	3	relu relu relu	32 22 11	adam	27.97
B	0 a 1	Yes	3	relu relu relu	32 22 11	adam	96.27
C	0 a 1	Yes	3	relu sigmoid sigmoid	57 48 36	adam	97.72
D	0 a 1	Yes	4	relu sigmoid sigmoid sigmoid	61 52 40 27	adam	99.12

The generalization behavior of the different MLP models in table 1, i.e., the ability to correctly recognize ADLs that had not been presented previously, based on the particular *dataset* data during training, can be reviewed in Fig 1.

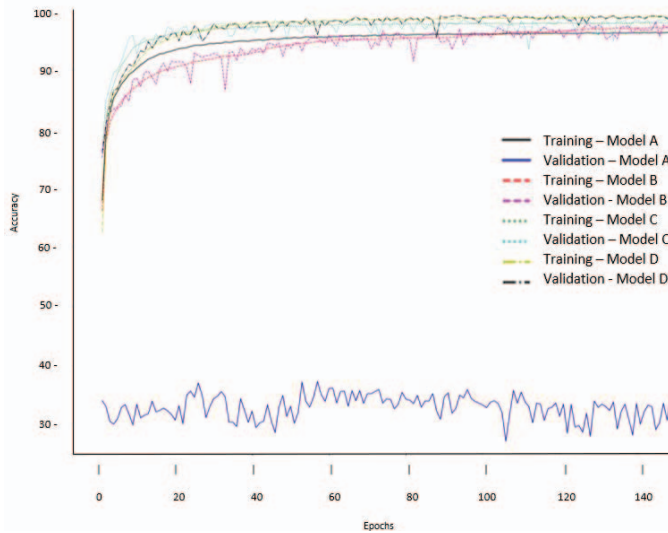


Fig. 1 Learning evolution and generalization capacity of the models described in table 1.

The Best MLP for *dataset* was the “Model D”. Using the cross-validation technique, this design yielded a mean accuracy of $99.03\% \pm 0.25\%$, which validates it as the best model using the complete database. Its architecture with details of the neurons for each layer is presented in Fig. 2.

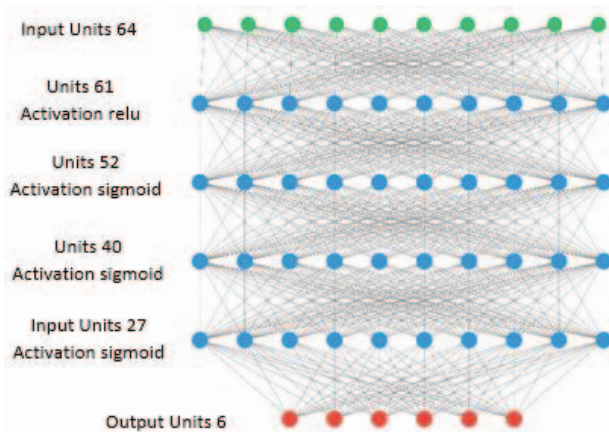


Fig. 2. Perceptron Multilayer's (MLP) architecture for Model D.

B. For the reduced database *rDB*

Table 2 shows the characteristics and adjustment parameters applied to the four main models where *rDB* was used, as well as the accuracy achieved and shown graphically in Fig. 3. “Model D-13” obtained within its execution the highest accuracy of 80% in the test process.

TABLE 2
PARAMETERS FOR MODELS A13 - D13

Model	Norm.	Random	# layers	A.F.	# neurons	Opt.	Accuracy %
A-13	0 a 1	No	2	relu sigmoid	10 8	adam	36.83
B-13	0 a 1	No	2	relu sigmoid	21 15	adam	34.83
C-13	0 a 1	Yes	4	relu relu sigmoid sigmoid	11 9 7 5	adam	70.17
D-13	-1 a 1	Yes	4	relu tanh tanh tanh	11 9 7 5	adam	80.70

Fig 3 shows the generalization behavior graph for the models presented in Table 2; where ADLs were recognized correctly without having been presented before to the ANN. This figure allows us to appreciate how, through the different modifications in the parameters, the behavior of the models was improved to reach an acceptable learning by applying dimensionality reduction to the entries in the database.

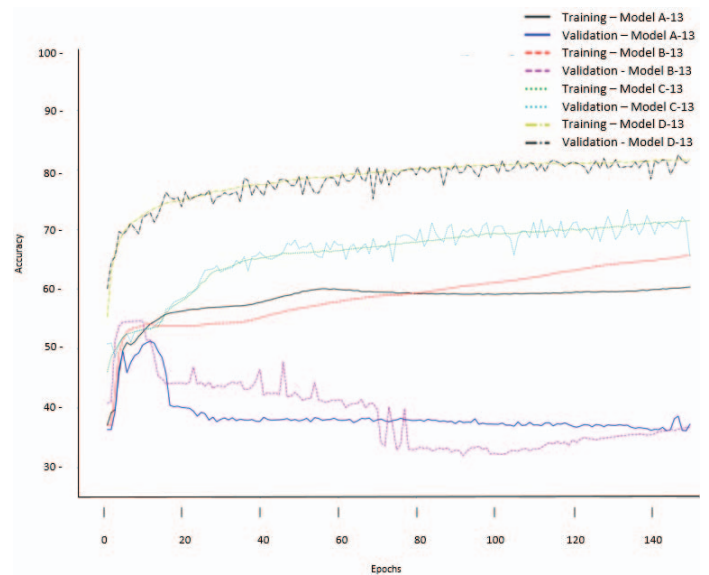


Fig. 3. Learning performance for MLP models using *rDB*

The best MLP for *rDB* achieved a learning rate of 80.7% at the testing stage. Its architecture with details of number of neurons for each layer is presented in Fig 4.

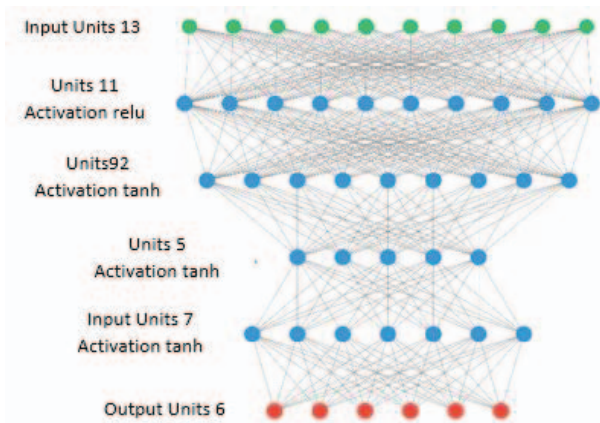


Fig. 4. Perceptron Multilayer's (MLP) architecture for "Model D-13".

The classification of activities performed by patients with PD has been studied using other techniques and algorithms with accuracy rates considered acceptable through Matlab programming. Previously, 8 grading methods were tested. Fig 5 is a graphical representation of the percentage of accuracy obtained for 6 classifiers tested [12].

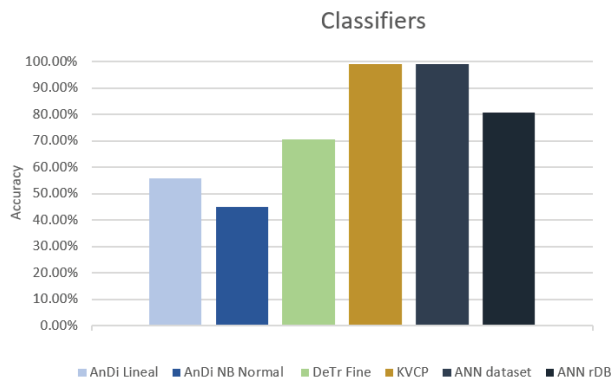


Fig 5. Accuracy of classifiers Linear Discriminant Analysis (AnDI Linear), Naïve Bayes Normal Discriminant Analysis (AnDI NB Normal), Fine Decision Tree (DeTr Fine), Weighted K-Near Neighbours (WKNN), ANN with *dataset* (ANN dataset) ANN with *rDB* (ANN rDB) used for ADL of patients with PD.

The last two methods used with ANN in this study were added to this graph. As it can be seen, a percentage of accuracy was obtained and considered to be increasingly acceptable. The time spent for training and validation was around 3 hours. However, it can be noted that the classification with ANN using *rDB* manages to be positioned at a higher position than other classifiers that used discriminant analysis and decision trees previously tested.

IV. DISCUSSION

The process of changing the number of layers, activation functions and number of neurons and normalization was appropriate to improve the percentage of learning for ANN in the classification of ADLs to reach 99% of accuracy. As

can be seen in Figure 1, the percentage of learning increases after those modifications were made to "Model A", as well as a stabilization in the behavior of the MLP after the 60th epoch of training and testing. Similarly, this method achieves results very similar to the KCVF classification method; this is reliable to help the specialist to a more accurate assessment in the monitoring and quality of rehabilitation in patients with PD.

A decrease in *dataset's* dimensionality to 21. 5% allowed us to have an appropriate classification learning for ANN without requiring so much input data and even surpassing accuracy percentages of other reported classification methods. This first attempt can help to reduce costs, problems of acquisition, processing and discomfort for users. On the other hand, developing the classification using artificial neural networks with Python and specialized libraries was shown to have the advantage of more user-friendly programming compared to *Matlab* for a user not expert in Artificial Intelligence or Data Science. Another advantage of this type of classification was the use of free-to-use programs without the need for any license for their application, compared to the classifiers previously used through Matlab.

E. Recommendations

After accomplishing the main objective of this study, through ANN programming for the ADLs classification of patients with PD, it remains possible to face several future studies; One of these, to improve the results obtained through tests using all the data available in the database, per subject, i. e. , to use data from a single patient in the training and thus be able to compare with the signals of another patient in the test stage. In this way, the reliability of the classification would be strengthened, especially for personalized treatments. Another possible future work would be to apply recurrent networks for use in real-time motion predictions during the execution of activities to evaluate performance and contribute to electrical stimulation rehabilitation. To continue the implementation and improvement of the work, it is proposed to analyze the results obtained from the data capture using less than 4 sensors at 50 Hz to improve practicality. However, to achieve a better learning rate, it would be suggested to increase the number of IMUs applied in the body to cover more reading areas and thus increase the amount of data input to the network. Similarly, it would be recommended to obtain a greater number of predictors through PCA and try to exceed 80% accuracy. Another thing to take in consideration would be to discuss with doctors what would be the minimum number of parameters they have to measure to establish a diagnosis or a follow-up of patients and see if the development of the model works with those parameters. As same as to discussion with doctors how serious would be a false-negative or a false positive classification. So we can tune the ANN models to fetch the best performance.

V. CONCLUSION

For the configuration of Artificial Neural Networks, the normalization and randomization of data is essential to obtain a good percentage of learning. For this work, the *Adam* optimizer turned out to provide better results than using the *sgd* optimizer. Similarly, it was concluded that, for the original database, using a four-layer ANN with *relu* and *sigmoid* activation functions presented a better performance than using a single type of AF and fewer layers. Using a database with only 21.5% of the original database (the most important signals applying PCA), satisfactory results can be obtained by resorting to a normalization of -1 to 1 and using the hyperbolic tangent activation function. With this, it is more practical to have fewer sensors therefore, the interest of classifying activities with fewer signals.

The main objective of this study was to explore through deep learning networks the classification of physical activities from a database of movements captured with inertial sensors. This objective has been achieved with a result close to the highest recorded by other classification techniques used for the database provided. The study developed showed the potential of applying Deep Learning and ANN to the resolution of classification problems to be later integrated into a system that supports medical staff in the monitoring of activities performed by patients with Parkinson's Disease.

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